Example 1: Show that the planes are parallel and distinct.

$\pi_1 : x+2y+3z-6=0 \qquad \pi_2 : 4x+8y+12z-25=0$

$\vec{n}_1 = [1, 2, 3]$

$\vec{n}_2 = [4, 8, 12]$

$\therefore \vec{n}_2 = 4\vec{n}_1 \Rightarrow$ parallel.

$D_1 = -6$

$D_2 = -25$

$\Rightarrow D_2 \neq 4D_1 \therefore$ distinct

Example 2: Show that the planes are parallel and identical.

$\pi_1 : 3x+2y+5z=4 \qquad \pi_2 : 9x+6y+15z-12=0$

$\vec{n}_1 = [3, 2, 5]$

$\vec{n}_2 = [9, 6, 15]$

$\therefore \vec{n}_2 = 3\vec{n}_1 \Rightarrow$ parallel

$D_1 = -4$

$D_2 = -12$

$\Rightarrow D_2 = 3D_1 \therefore$ identical (coincident)